# The High-Dimensional Asymptotics of Principal Component Regression

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# Quick review for PCR

- PCR: Combine OLS regression + PCA:

$$\widehat{\beta}_{\mathrm{PCR}}(m) = (X_m^\top)^\dagger y$$

where $X_m \in \mathbb{R}^{n\times p}$ retains $m$ largest PCs of $X$.

$$X \overset{\mathrm{SVD}}{=} U\,\mathrm{diag}(\sigma_1, \ldots, \sigma_r) W^\top,$$

$$X_m = U\,\mathrm{diag}(\sigma_1, \ldots, \sigma_m, 0, \ldots, 0) W^\top.$$

Note: $m = r$ coincides with OLS.

- Denote $S = \frac{1}{n} X^\top X$ the sample covariance.

$$P_m = \text{proj. on largest } m \text{ eigspaces of } S.$$

$$P_m^\perp = \text{proj. on orth. complement.}$$

$$(\text{So } X_m = XP_m.)$$

# Motivation: Why PCR in High Dimensions?

- **Classical View ($n \gg p$):** PCR reduces variance by projecting onto high-variability directions. Sample PCs $\approx$ Population PCs.
- **The High-Dimensional Challenge ($p/n \to \gamma > 0$):** Sample eigenvectors become **inconsistent**.
- **The Core Question:** How does high-dimensional noise distort PCR, and exactly how many PCs ($m$) should we retain to optimize the Bias-Variance tradeoff?

**Key Insight**

High-dimensionality "smears" the true signal across empirical eigenvectors. We need Random Matrix Theory (RMT) to precisely quantify this signal leakage.

# Problem Setup

- **Linear Model:** $y = X\beta^* + \xi$, where $\xi \sim \mathcal{N}(0, \sigma^2 I)$.
- **Covariates:** $X = Z\Sigma^{1/2}$, with $Z_{ij}$ i.i.d. (mean 0, var 1).
- **PCR Estimator:** $\hat{\beta}_{PCR} = (\mathcal{P}_m X^\top)^\dagger y$, where $\mathcal{P}_m$ projects onto the $m$ leading *sample* PCs.
- High-dimensional asymptotics:

$$p = p_n, \quad m = m_n, \qquad n, p_n, m_n \to \infty,$$

$$\frac{p_n}{n} \to \gamma \in (0, \infty), \qquad \frac{m_n}{p} \to \alpha \in [0, \rho^S].$$

- $\rho^S := \lim_{n\to\infty} \frac{\text{rank}(S)}{p}$.

# Problem Setup

- Has LSD. Denote ESD:

$$\hat{H}_n(\tau) = \frac{1}{p}\sum_{i=1}^{p} \mathbf{1}_{\lambda_i(\Sigma)\leq\tau}.$$

  Then

$$\hat{H}_n \Longrightarrow H,$$

  for *compactly supported* $H$.

- Finitely many *outliers* at upper edge:

$$\lambda_1(\Sigma) > \cdots > \lambda_{k_0}(\Sigma) > \max \operatorname{supp}(H)$$

  fixed, while

$$\max_{k_0<i\leq p} \operatorname{dist}\big(\lambda_i(\Sigma), \operatorname{supp}(H)\big) \longrightarrow 0.$$

(Spiked covariance w/ pop profile $H$.)

# Sample Covariance: Quick Recap

- Sample ESD:

$$\hat{F}_n(\theta) = \frac{1}{p} \sum_{i=1}^{p} \mathbf{1}_{\lambda_i(S) \le \theta}$$

  converges to Marchenko–Pastur law:

$$\hat{F}_n \Longrightarrow F_{\gamma,H}.$$

  Consists of: density $f_{\gamma,H}(\theta)$ with compact support $\mathcal{S}$; atom at zero ($S$ rank-deficient).
- Finitely many outliers. There is $\tau_{\mathrm{BBP}} > \tau^+$ s.t.: Eigenval. $1 \le i \le k_0$ is an outlier if

$$\lambda_i(\Sigma) > \tau_{\mathrm{BBP}};$$

  if so, formula exists for outlier location $\vartheta_i$, similarly for population-sample eigenvector overlap.

  Denote $k_0^+$ number of outliers; $\mathcal{O}^+ = \{\vartheta_1 > \cdots > \vartheta_{k_0^+}\}$.

  Other eigvals are non-outliers:

$$\lim_{n\to\infty} \max_{k_0^+ < i \le p} \operatorname{dist}\big(\lambda_i(S), \mathcal{S}\big) \to 0.$$

# Problem Setup

**Signal Alignment (Spectral Measure $G$)**
The "geometry" of the problem is captured by how the signal $\boldsymbol{\beta}^*$ is distributed across the eigenspaces of the population covariance $\boldsymbol{\Sigma}$:

$$\hat{G}_n(\tau) = \frac{1}{\|\boldsymbol{\beta}^*\|^2} \sum_{i=1}^{p} \langle \boldsymbol{\beta}^*, \boldsymbol{v}_{i,n} \rangle^2 \mathbf{1}_{\{\tau_{i,n} \leq \tau\}} \xrightarrow{a.s.} G(\tau)$$

*$G$ describes whether the signal is concentrated in high-variance ($\tau$ large) or low-variance ($\tau$ small) population directions.*

# Main Theorem I: Asymptotic Estimation Risk

- Defined as
$$R_n^{\mathrm{Est}}(\beta^\star, X) = \mathbb{E}\left[\left\|\hat{\beta}_{\mathrm{PCR}} - \beta^\star\right\|^2 \,\middle|\, \beta^\star, X\right]$$
"How well do we recover the model parameters?"
- *Bias–variance* decomposition
$$R_n^{\mathrm{Est}}(\beta^\star, X) = B_n^{\mathrm{Est}}(\beta^\star, X) + \sigma^2 V_n^{\mathrm{Est}}(X)$$
where
$$B_n^{\mathrm{Est}}(\beta^\star, X) = \left\|P_m^\perp \beta^\star\right\|^2,$$
$$V_n^{\mathrm{Est}}(X) = \frac{1}{n}\operatorname{tr}\big((P_m S P_m)^\dagger\big).$$
- *Interpretation:*
We perform OLS regression on $\mathrm{Range}(P_m)$;
*Bias:* We unavoidably miss part of $\beta^\star$ on the complement;
*Variance:* Suffer only the part of noise on $\mathrm{Range}(P_m)$.

# Main Theorem I: Asymptotic Estimation Risk

**Theorem.**

Almost surely as $n, m, p \to \infty$,

$$V_n^{\mathrm{Est}}(\beta^\star, X) \;\to\; V_\infty^{\mathrm{Est}},$$

$$B_n^{\mathrm{Est}}(\beta^\star, X) \;\to\; b_{\mathrm{Opt}}^{\mathrm{Pred}} + b_{\mathrm{Bulk}}^{\mathrm{Est}}(\alpha).$$

*Interpretation:*

- Variance (Noise Amplification): $V_\infty^{Est}(\alpha) = \gamma \int_{Q^{-1}(\alpha)}^{\theta^+} \frac{1}{\theta} dF_{\gamma,H}(\theta)$.
  *(Integral of inverse eigenvalues over the retained Marchenko-Pastur spectrum).*
- $b_{\mathrm{Opt}}^{\mathrm{Est}}$ unavoidable bias,
  $$\left\| \beta^\star - P^{\perp}_{\mathrm{rank}(S)} \beta^\star \right\|^2.$$
- $b_{\mathrm{Bulk}}^{\mathrm{Est}}(\alpha)$ is bias associated with discarding bulk sample PCs.
  Each discarded PC contributes $O(1/p)$ bias.

# Main Theorem II: (Out-of-Sample) Prediction Risk

- Defined as

$$R_n^{\mathrm{Pred}}(\beta^\star, X) = \mathbb{E}\left[\left(y_{\mathsf{new}} - x_{\mathsf{new}}^\top \hat{\beta}_{\mathrm{PCR}}\right)^2 \,\middle|\, \beta^\star, X\right] - \sigma^2$$

$$= \mathbb{E}\left[\left\|\Sigma^{1/2}(\beta^\star - \hat{\beta}_{\mathrm{PCR}})\right\|^2 \,\middle|\, \beta^\star, X\right].$$

where $x_{\mathsf{new}} \sim N(0, \Sigma)$, $y_{\mathsf{new}} = x_{\mathsf{new}}^\top \beta^\star + \eta_{\mathsf{new}}$.
"How well can we predict the label of a new data point?"

- *Bias–variance* decomposition

$$R_n^{\mathrm{Pred}}(\beta^\star, X) = B_n^{\mathrm{Pred}}(\beta^\star, X) + \sigma^2 V_n^{\mathrm{Pred}}(X)$$

where

$$B_n^{\mathrm{Pred}}(\beta^\star, X) = \left\|\Sigma^{1/2} P_m^\perp \beta^\star\right\|^2,$$

$$V_n^{\mathrm{Pred}}(X) = \frac{1}{n}\,\mathsf{tr}\big(\Sigma(P_m S P_m)^\dagger\big).$$

# Main Theorem II: (Out-of-Sample) Prediction Risk

**Theorem.**

Almost surely as $n, m, p \to \infty$,

$$V_n^{\mathrm{Pred}}(\beta^\star, X) \;\to\; V_\infty^{\mathrm{Pred}},$$

$$B_n^{\mathrm{Pred}}(\beta^\star, X) \;\to\; b_{\mathrm{Opt}}^{\mathrm{Pred}} + b_{\mathrm{Bulk}}^{\mathrm{Pred}}(\alpha).$$

# Case Study: Isotropic Prior

- Simplest case: $\beta^\star \sim \mathrm{Unif}(\mathbb{S}^{p-1}),\ G = H$.
- Formulas simplify *considerably*:

$$R_\infty^{\mathrm{Est}} = \underbrace{1-\alpha}_{\text{Bias}} + \underbrace{\sigma^2\gamma \int_{Q^{-1}(\alpha)}^{\theta^+} \frac{1}{\theta} f_{\gamma,H}(\theta)\, d\theta}_{\text{Variance}}.$$

and

$$R_\infty^{\mathrm{Pred}} = \underbrace{\frac{1}{\gamma\, m(0)} \cdot \mathbf{1}_{\rho^S < \rho^\Sigma} + \int_0^{Q^{-1}(\alpha)} \frac{1}{\theta\, |m(\theta)|^2} f_{\gamma,H}(\theta)\, d\theta}_{\text{Bias}}$$

$$\underbrace{+\, \sigma^2\gamma \int_{Q^{-1}(\alpha)}^{\theta^+} \frac{1}{\theta^2\, |m(\theta)|^2} f_{\gamma,H}(\theta)\, d\theta}_{\text{Variance}}.$$

# Case Study: Isotropic Prior

- Can solve for the optimal $\alpha^\star$:

$$\alpha^\star = Q\left(\frac{\gamma}{\mathrm{SNR}}\right), \qquad \mathrm{SNR} := \frac{1}{\sigma^2}.$$

- In particular:
  - If $\frac{\gamma}{\mathrm{SNR}} \geq \theta_+$, then $\alpha^\star = 0$ (low-SNR; discard all PCs)
  - If $\frac{\gamma}{\mathrm{SNR}} \leq \theta_-$, then $\alpha^\star = \rho^S$ (high-SNR; take all PCs)
  - If $\theta_- < \frac{\gamma}{\mathrm{SNR}} < \theta_+$, then $0 < \alpha^\star < \rho^S$.
- Special case: $\Sigma = I$. Then

$$f_\gamma(\theta) = \frac{1}{2\pi\gamma\theta}\sqrt{(\theta_+ - \theta)(\theta - \theta_-)}, \qquad \theta_\pm = (1 \pm \sqrt{\gamma})^2.$$

# Case Study: Isotropic Prior

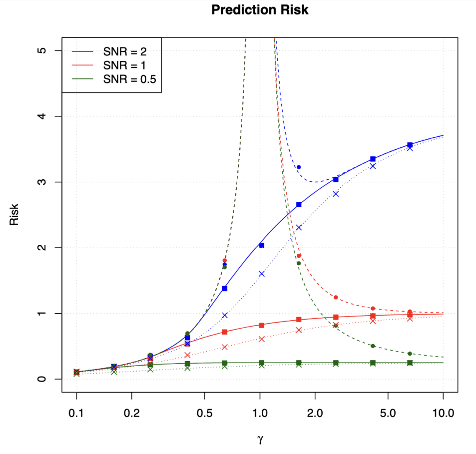


**Figure 1:** Prediction risk for different values of SNR. Solid: PCR; dotted with crosses: Ridge; dotted with points: OLS.

# Case Study: Spiked Covariance

- $\Sigma = \mathrm{diag}(\tau, 1, \ldots, 1), \quad \tau > 1;\ \beta^\star = e_1$. Simplest model of *low-dimensional latent structure*.
- Formulas simplify:

$$R_\infty^{\mathrm{Est}} = \underbrace{\frac{\gamma - 1}{\gamma - 1 + \tau_1}\mathbf{1}_{1/\gamma<1} + \int_0^{Q_\gamma^{-1}(\alpha)} \frac{\gamma\tau_1}{\tau_1^2 - \tau_1(1-\gamma+\theta)+\theta} f_\gamma(\theta)\, d\theta}_{\text{Bias}}$$

$$+ \underbrace{\int_{Q_\gamma^{-1}(\alpha)}^{\theta_\gamma^+} \frac{1}{\theta} f_\gamma(\theta)\, d\theta}_{\text{Variance}}.$$

and

$$B^{\mathrm{Pred}} = (\tau_1 - 1)\big(B^{\mathrm{Est}}\big)^2 + B^{\mathrm{Est}}, \qquad V^{\mathrm{Pred}} = V^{\mathrm{Est}}.$$

# Case Study: Spiked Covariance

- Note: $\alpha^\star$ for estimation $\neq \alpha^\star$ for prediction!
- For estimation risk, easy to find $\alpha^\star$:

$$\alpha^\star = Q_\gamma\big(\chi(\tau_1, \mathrm{SNR})\big), \qquad \chi(\tau_1, \mathrm{SNR}) = \frac{\gamma\tau_1 + \tau_1(\tau_1 - 1)}{\mathrm{SNR} + \tau_1 - 1}.$$

- In particular:
  - If $\chi \geq (1+\sqrt{\gamma})^2$, then $\alpha^\star = 0$ (low-SNR; discard all PCs)
  - If $\chi \leq (1-\sqrt{\gamma})^2$, then $\alpha^\star = \rho^S$ (high-SNR; take all PCs)
  - If $(1-\sqrt{\gamma})^2 < \chi < (1+\sqrt{\gamma})^2$, then $0 < \alpha^\star < \rho^S$.

- *Importantly:* There is a regime (depending on $\tau_1$, SNR) where one sees a strong sample outlying eigenvalue, but *still*, better to take a large fraction of sample PCs.

# Case Study: Spiked Covariance

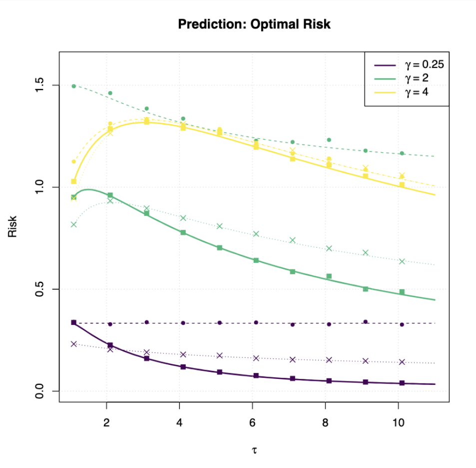


**Figure 2:** Risk of optimally tuned PCR. Solid: PCR; dotted with crosses: Ridge; dotted with points: OLS.

# Summary & Takeaways

1. **PCR in high dimensions behaves fundamentally differently**
    - Sample PCs are *noisy* and misaligned with population directions
    - Signal is **spread across many empirical eigenvectors**
2. **Risk is governed by spectral geometry**
    - Bias: missing signal outside selected PCs
    - Variance: amplification from small eigenvalues
    - Tradeoff depends on **spectrum + signal alignment**
3. **Optimal number of PCs is nontrivial**
    - Not always: "keep top PCs"
    - Depends on: SNR; aspect ratio $\gamma = p/n$; signal alignment with spectrum.

**Key takeaway:**

*PCR is not just dimension reduction — its performance is entirely determined by* **how signal aligns with the random spectrum**.

# High-dimensional Analysis of Synthetic Data Selection

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# Motivation

**Synthetic data for machine learning**

Synthetic data can address difficulties of observational data: privacy, class imbalance, and data scarcity.

**But the empirical picture is murky**

- Results vary across datasets and architectures.
- Heuristic: "keep synthetic data close to real" — but *close in which sense?*
- No rigorous theory pinpoints which distributional properties drive generalisation.

**The Research Question**

Which properties of the synthetic distribution affect generalisation error?

- **Mean shift?** $\mu_s \neq \mu_t$
- **Covariance shift?** $\Sigma_s \neq \Sigma_t$

**Main Question**

How does extra synthetic data $(X_s, y_s)$ from a generative model help training, and how should we *select* it to minimise test error?

# Setting

## Real (target) data

- $n_t$ samples, $p$ features
- $y_t = X_t\beta + \varepsilon_t$
- Rows of $X_t \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_t, \Sigma_t)$
- $\varepsilon_t$ i.i.d., mean 0, variance $\sigma^2$
- $n_t$ typically *small*

## Synthetic data

- $n_s$ samples from generative model
- $y_s = X_s\beta + \varepsilon_s$
- Rows of $X_s \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_s, \Sigma_s)$
- $\varepsilon_s$ i.i.d., mean 0, variance $\sigma^2$
- $n_s$ can be *large*

## High-dimensional regime

$$p,\ n_t,\ n_s \to \infty, \qquad \frac{n_t}{p} \to \kappa_t,$$

$$\frac{n_s}{p} \to \kappa_s$$

(fixed ratios)

## Estimator

Ridgeless least squares on the combined dataset $[X_t;\ X_s]$

# Main Theoretical Result

**Theorem (Theorem 4.1 — informal)**

*For ridgeless least squares trained on real data augmented with synthetic data, in the proportional high-dimensional limit ($p, n_t, n_s \to \infty$ at fixed rates), the asymptotic generalisation error depends on the* **covariance** $\Sigma_s$ *of the synthetic distribution but is* **independent of the mean** $\mu_s$. *Moreover, in natural settings, the choice $\Sigma_s = \Sigma_t$ (covariance matching) is optimal.*

**× Mean shift is irrelevant**

$\mu_s \neq \mu_t$ does *not* change the asymptotic generalisation error. This is surprising: intuition suggests "mean closer to real = better," but the theory says otherwise.

**✓ Covariance shift is critical**

$\Sigma_s \neq \Sigma_t$ directly controls the generalisation error. Furthermore, $\Sigma_s = \Sigma_t$ is *provably optimal* in several settings ⇒ motivates covariance matching.

# Proof Sketch — Setting Up the Risk

## Data model (eq. 3.2)

Both datasets share the structure $X_{(i)} = Z^{(i)}\Sigma_{(i)}^{1/2} + \mathbf{1}_{n_{(i)}}\mu_{(i)}^\top$, where $Z^{(i)}$ has i.i.d. zero-mean, unit-variance entries. The pooled matrix is

$$X = \begin{bmatrix} X_t \\ X_s \end{bmatrix} \in \mathbb{R}^{n\times p}, \quad n = n_t + n_s, \qquad \hat{\beta} = (X^\top X)^+ X^\top y \quad \text{(min-norm interpolator)}.$$

## Excess risk decomposition (eq. 3.3 – 3.5)

Testing on the *same* distribution as $(X_t, y_t)$ (non-zero mean!), the excess risk is

$$R_X(\hat{\beta};\beta) = \underbrace{\beta^\top \Pi(\Sigma_t + \mu_t\mu_t^\top)\Pi\beta}_{B_X = 0 \text{ (under-param.)}} + \underbrace{\frac{\sigma^2}{n}\operatorname{Tr}\left[\hat{\Sigma}^+(\Sigma_t + \mu_t\mu_t^\top)\right]}_{V_X},$$

where $\hat{\Sigma} = X^\top X/n$ and $\Pi = I - \hat{\Sigma}^+\hat{\Sigma}$. In the **under-parameterized regime** ($n > p$), $\hat{\Sigma}$ is full rank so $\Pi = 0$, meaning *bias vanishes* and only the variance $V_X$ needs to be characterized.

# Proof Sketch — Why the Mean Doesn't Matter

## Factoring out the means as a rank-2 perturbation

Using the data model $X_{(i)} = Z^{(i)}\Sigma_{(i)}^{1/2} + \mathbf{1}_{n_{(i)}}\mu_{(i)}^\top$, the sample covariance satisfies

$$\hat{\Sigma} = \frac{X^\top X}{n} = \underbrace{\frac{1}{n}\sum_{i\in\{t,s\}} \Sigma_{(i)}^{1/2} Z^{(i)\top} Z^{(i)} \Sigma_{(i)}^{1/2}}_{\text{zero-centered part}} + \underbrace{\frac{1}{n}\left(n_t\, \mu_t\mu_t^\top + n_s\, \mu_s\mu_s^\top\right)}_{\text{rank-}\leq 2 \text{ perturbation}}.$$

The mean contribution is a **rank-$\leq$2 perturbation** (Props. A.1–A.3).

## Anisotropic local laws (RMT)

For the zero-centered part, anisotropic local laws give a deterministic equivalent for $\mathrm{Tr}[\hat{\Sigma}^{+}(\Sigma_t + \mu_t\mu_t^\top)]$. finite-rank perturbations leave the *empirical spectral distribution* of $\hat{\Sigma}$ unchanged in the limit $p \to \infty$ — so the rank-2 mean terms **drop out entirely**.

## Consequence (Theorem 4.1)

The asymptotic excess risk depends only on the eigenvalues of $M^\top M$, where $M = \Sigma_s^{1/2}\Sigma_t^{-1/2}$, and **not** on $\mu_t$ or $\mu_s$. Convergence rate: $O(\sigma^2 p^{-1/2})$.

# Proof Sketch — The Exact Formula & Covariance Optimality

## Theorem 4.1 — exact asymptotic formula (under-parameterized)

Let $M = \Sigma_s^{1/2}\Sigma_t^{-1/2}$ with eigenvalues $\lambda_1 \geq \cdots \geq \lambda_p$ of $M^\top M$. Then, with high probability,

$$\lim_{n\to\infty} \left| R_X(\hat{\beta};\beta) - \frac{\sigma^2}{n}\operatorname{Tr}\left[(\alpha_1 M^\top M + \alpha_2 I_p)^{-1}\right] \right| = 0,$$

where $\alpha_1, \alpha_2 > 0$ are the *unique* solutions to the self-consistent equations

$$\alpha_1 + \alpha_2 = 1 - \frac{p}{n}, \qquad \alpha_1 + \frac{1}{n}\sum_{i=1}^{p} \frac{\lambda_i \alpha_1}{\lambda_i \alpha_1 + \alpha_2} = \frac{n_s}{n}.$$

## ✓ Covariance matching is optimal (Thm. 4.3)

The risk depends on $\Sigma_s$ only through the eigenvalues of $M^\top M = \Sigma_s \Sigma_t^{-1}$. Minimising over all PSD $\Sigma_s$ yields $\Sigma_s^* \propto \Sigma_t$ (i.e., $M^\top M = cI$ for some scalar $c > 0$), proved via a convexity argument. Setting $c = 1$ ($\Sigma_s = \Sigma_t$) further minimises over $c$.

## Why $M^\top M$ is the right object

$M^\top M = \Sigma_t^{-1/2}\Sigma_s\Sigma_t^{-1/2}$ measures the *relative* spectral alignment of $\Sigma_s$ w.r.t. $\Sigma_t$. When $\Sigma_s = \Sigma_t$: all eigenvalues of $M^\top M$ equal 1, the formula reduces to $\frac{\sigma^2 p}{n(\alpha_1+\alpha_2)}$, the minimum.

# Proof Roadmap — Summary

| | | | |
|---|---|---|---|
| **1** Bias–variance decomposition $B_X = 0$, analyze $V_X$ | **2** Factor means as rank-2 perturbation (Props. A.1–A.3) | **3** Anisotropic local laws (RMT) for zero-centered part | **4** Self-consistent eqs. for $\alpha_1, \alpha_2$; optimise over $\Sigma_s$ |

*Technical backbone: Anisotropic local laws for sample covariance matrices · Finite-rank perturbation theory (RMT) · Self-consistent fixed-point equations · Convexity argument for optimality*

### × Mean-invariance

Both $\mu_t$ and $\mu_s$ appear only as a *rank-*$\leq 2$ additive perturbation to $\hat{\Sigma}$. By RMT, finite-rank perturbations are invisible to the ESD — so the asymptotic risk is independent of $\mu_t, \mu_s$.

### ✓ Covariance optimality

Risk depends on $\Sigma_s$ only through eigenvalues of $M^\top M = \Sigma_t^{-1/2} \Sigma_s \Sigma_t^{-1/2}$. Convexity $\Rightarrow$ optimum at $M^\top M = I$, i.e., $\Sigma_s^* = \Sigma_t$.

# Conclusions

1. **Rigorous theory.** First high-dimensional analysis identifying *covariance* (not mean) as the relevant property of synthetic data for generalisation error.
2. **Proof techniques.** GET + RMT deterministic equivalents $\Rightarrow$ exact closed-form error formula $\Rightarrow$ mean-invariance via rank-2 perturbation argument $\Rightarrow$ variational optimality of $\Sigma_s = \Sigma_t$.
3. **Covariance matching.** A principled, theory-grounded selection algorithm that consistently outperforms existing baselines across diverse experimental settings.

**Practical guideline**

When selecting synthetic samples for augmentation, **match the feature-space covariance** to the target distribution. Mean alignment alone is insufficient and it doesn't matter.

# *A Universal Trade-off Between the Model Size, Test Loss, and Training Loss of Linear Predictors (2023)*

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*Authors: Nikhil Ghosh and Mikhail Belkin*

April 28, 2026

# Motivation

**Two Frameworks:**

- *Classical statistics* tries to approximate test loss with training loss (training loss close to noise level).
- *Modern deep learning* obtains a much lower loss on the training set than on the test set, often via overparameterization.
  - *Overparameterization* is when there are many more features included than is needed to fit the training data exactly.
  - Let $n$ denote the number of training observations and $p$ denote the number of parameters.
  - Overparameterization has $p \gg n$.
  - *Benign overfitting* is the phenomenon in which overparameterized models still generalize near optimally.

**Central question**

What is the relationship between training (empirical) loss, test (expected) loss, and the number of parameters $p$ relative to the sample size $n$?

# Setting

- **Data:** $D_n = \{(x_i, y_i)\}_{i=1}^n \sim_{\text{iid}} P$ on $\mathbb{R}^d \times \mathbb{R}$.
- **Losses:**

$$\widehat{\mathcal{R}}_n(f) = \frac{1}{n} \sum_i (y_i - f(x_i))^2 \quad \text{(training)}$$

$$\mathcal{R}(f) = \mathbb{E}_{(x,y)}(y - f(x))^2 \quad \text{(test)}$$

- **Excess loss:** $\mathcal{E}(f) = \mathcal{R}(f) - \mathcal{R}(f_\star)$, where $f_\star(x) = \mathbb{E}[y \mid x]$ is the *regression function*.
  - Note that $f_\star(x) = \operatorname{argmin}_f R(f)$.
- Assume noise level of at least $\sigma^2$:

$$\operatorname{Var}(y \mid x) \geq \sigma^2 \ a.s.$$

# Setting

- We consider linear feature models, of the form

$$\beta(\mathbf{x}) = \beta^T \phi_p(\mathbf{x}),\ \beta \in \mathbb{R}^p,$$

  where $\phi_p$ is a $p$-dimensional feature map: $\phi_p : \mathbb{R}^d \to \mathbb{R}^p$.
- $\beta_*$ is the optimal linear predictor, and $\mathcal{E}^{lin}(\beta) := R(\beta) - R(\beta_*)$ is the excess loss compared against the optimal linear predictor.
- We consider algorithms $\mathcal{A}$ which output linear predictors: $\mathcal{A}(D_n) = \beta$.

# The Key Parameter: $\tau$

For an algorithm $\mathcal{A}$, we measure training loss, normalized by noise level. We impose a bound on this value, $\tau$:

$$\frac{\widehat{\mathcal{R}}_n(\mathcal{A}(D_n))}{\sigma^2} \leq \tau.$$

**$\tau$ is a knob between two regimes:**

- $\tau = 0$: **interpolation**.
- $0 < \tau < 1$: fitting some noise but not all.
- $\tau = 1$: **classical regime** (training loss at noise level).
- $\tau > 1$: underfitting.

# Main Result: Theorem 1

## Theorem 1 (Universal Lower Bound)

Let $n, p \geq 1$. Assume mild non-degeneracy ($\mathrm{rank}(\Phi) = \min(n, p)$ a.s., $\mathbb{E}y^2 < \infty$, $\mathrm{Var}(y \mid x) \geq \sigma^2$).
For *any* algorithm $\mathcal{A}$ outputting a linear predictor with $\widehat{\mathcal{R}}_n(\mathcal{A}(D_n)) \leq \tau\sigma^2$ a.s.:

$$\mathbb{E}_{D_n \sim P^n}\mathcal{E}(\mathcal{A}(D_n)) \;\geq\; \sigma^2 \cdot \frac{n}{p} \cdot (1 - \sqrt{\tau})^2.$$

**Three observations:**

1. **Algorithm-independent** — holds even for *oracle algorithms* (always output $\beta_\star$).
2. **Distribution-independent** — minimal assumptions on the distribution $P$ from which the data is sampled.
3. **Non-asymptotic** — holds for finite $n$ and $p$.

# Interpreting the Bound

$$\mathbb{E}_{D_n \sim P^n} \mathcal{E}(\mathcal{A}(D_n)) \;\geq\; \sigma^2 \cdot \frac{n}{p} \cdot (1 - \sqrt{\tau})^2$$

**To make excess loss small, you need either:**

- $\tau \to 1$ (classical: don't try to fit below the noise level), **or**
- $p/n \to \infty$ (modern: overparameterize heavily).

**Two cases to focus on:**

- **Peak** ($p \approx n$): bound loose; behavior depends sharply on $\tau$.
- **Tail** ($p \gg n$): bound tight; behavior robust to $\tau$.

**Consequence for convergence rates.** To have excess loss converge to 0 as $n \to \infty$ as $\mathcal{O}(n^{-\alpha})$ with $\tau$ bounded from 1, we need:

$$p = \Omega(n^{1+\alpha}).$$

($n^\alpha$ *times* as many parameters as it takes to interpolate)

# Proof Sketch of Theorem 1

**Step 1: Reformulate.** Setting $\mathcal{E}_\star(\tau; n, p) = \min_\beta \mathcal{E}(\beta)$ $s.t.$ $\hat{R}_n(\beta) \leq \tau\sigma^2$, we find that it suffices to bound $\mathbb{E}[\mathcal{E}_\star(\tau; n, p)]$ instead. We have $\mathcal{E}_\star \geq \mathcal{E}_\star^{lin}$, and so we work with

$$\mathcal{E}_\star^{lin}(\tau) = \min_{b \in \mathbb{R}^p} \|b\|_2^2 \; s.t. \; \frac{1}{n}\|Wb - \xi\|_2^2 \leq \tau\sigma^2.$$

($\xi$ is noise and $W$ is the whitened feature matrixt).

**Step 2: Convert to Unconstrained Problem.** Use Lagrangian to find that for any $\lambda \geq 0$:

$$\mathcal{E}_\star^{lin}(\tau) \geq \min_b \left\{ \|b\|^2 + \tfrac{n}{p\lambda}\left(\tfrac{1}{n}\|Wb - \xi\|^2 - \tau\sigma^2\right)\right\}.$$

**Step 3: The minimizer is ridge regression.** Solving the optimization problem for $b_\star$, we find

$$b_\star(\lambda) = \tfrac{1}{p}W^T\left(\lambda I + \tfrac{1}{p}WW^T\right)^{-1}\xi$$

— which is exactly the *ridge regression* estimator with features $W$, target $\xi$, and parameter $\lambda$.

**Step 4: Bound the Ridge Variance.** Choose $\lambda_\star$ so the constraint is tight in expectation. Then $\mathbb{E}\|b_\star(\lambda_\star)\|^2 \geq \sigma^2 \cdot (n/p) \cdot (1 - \sqrt{\tau})^2$ via Chebyshev's sum and Jensen inequalities.

# Marchenko-Pastur Asymptotics

Under MP assumptions (and well-specified linear model, $n/p \to \gamma$), the results are sharper:

- Exact asymptotic formula via the Stieltjes transform ($m$):

$$\mathcal{E}_\star(\tau,\gamma) = \sigma^2\,\gamma\,[m(-\lambda;\gamma) - \lambda m'(-\lambda;\gamma)], \quad \lambda^2 m'(-\lambda;\gamma) = \tau.$$

  ($\mathcal{E}_\star$ is minimum possible excess loss).
- **Three findings:**
  - **Theorem 2:** Universal bound is *tight* as $\gamma \to 0$ (heavy overparameterization).
  - **Theorem 4:** Sharper bound near the peak: $\mathcal{E}_\star \geq \frac{\sigma^2\gamma}{1-\gamma}\left(1 - \sqrt{\tau/(1-\gamma)}\right)^2$ for small $\tau \in [0, 1-\gamma]$ and $\gamma \in (0,1)$.
  - **Theorem 5:** At the peak ($\gamma = 1$): $\mathcal{E}_\star/\sigma^2 = \frac{1}{4\tau} + \frac{\tau}{4} - \frac{1}{2} = \Theta(\tau^{-1})$.

# Conclusion

**Universal trade-off:** Any near-optimal linear predictor is either classical ($\tau \to 1$) or heavily overparameterized ($p \gg n$).

**Under mild assumptions, a universal bound holds:**

- Distribution-free, algorithm-free, non-asymptotic.
- Tight in the regime $p \gg n$.

**Marchenko-Pastur Asymptotics:** In the MP setting, the asymptotic excess loss can be computed explicitly, providing insight into the strengths and limitations of the universal bound.